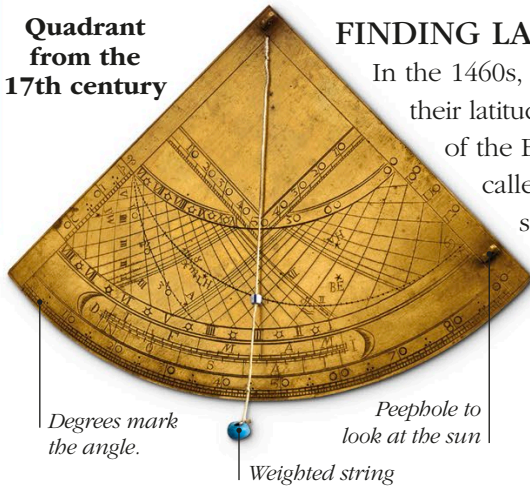


Navigating at sea

Sailors venturing out to sea need navigational aids to help them find their way. Over the centuries, instruments such as the quadrant and sextant were developed. These tools helped sailors determine their location from the height of the sun and the stars. Sailors also relied on the magnetic compass. Today, ships rely entirely on electronic satellite systems.

Quadrant from the 17th century



FINDING LATITUDE

In the 1460s, navigators began figuring out their latitude—their position north or south of the Equator—using an instrument called the quadrant. They would see the North Star at night, or the sun at midday, through two peepholes along an edge of the quadrant. Once the star and peepholes aligned, a weighted string showed the latitude on the dial.

An explorer uses a sextant in Antarctica, 1930



SEXTANT

In the 1730s, the Englishman John Hadley and American Thomas Godfrey invented the sextant, independent of each other. Looking into the sextant's telescope, a navigator could find out the angle of the sun and stars relative to the horizon by lining up two mirrors. Printed tables then gave the latitude for the angle. For years, this became the ultimate navigation tool.

These oscillating weights are not affected by the rolling motion of a ship on water.



► THE SEA CLOCK

John Harrison's marine chronometer solved the problem of timekeeping at sea.